

Beyond the Verdict: Learning Bayesian Risk Paths for LLM Agents

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Abstract

Safety failures of LLM agents unfold across turns: an external instruction is observed, adopted as a task, and converted into an action with a real consequence. A trajectory-level label can detect the outcome but cannot expose this local chain, while a post-hoc rationale need not reflect the computation that produced the verdict. We introduce TraceAudit, a 4,364-trajectory audit dataset with turn-level atomic facts, local relations, mechanism instances, and harm categories. We further introduce BayesTrace, a template-constrained differentiable Bayesian network in which each trajectory activates a local atomic–relation–mechanism–harm subgraph. We call its ranked, posterior-contributing routes *Bayesian risk paths*. Semantic relation calibration is separated from sample-dependent relation influence, allowing the relation shown in a risk path and its contribution to downstream mechanisms to remain explicit within the same graph. BayesTrace preserves risk-detection performance: the reported final checkpoint obtains 0.847 F1, while three fixed-protocol training seeds average 0.840 ± 0.006 F1, comparable to the strongest baseline at 0.843. On unsafe trajectories with path-content labels, it reaches 0.567 turn-evidence F1, 0.519 mechanism F1, and 0.563 harm-category F1. Deleting the displayed risk path moves the model’s own posterior $5.08\times$ more than deleting non-path turns. These results show that a constrained probabilistic structure can make readable path variables part of the prediction rather than text generated after it.

1 Introduction

LLM agents retrieve messages, read files, browse, call APIs, execute code, and change external state. These interactions break the assumptions of single-turn refusal evaluation: a trajectory can be unsafe because the agent executed an instruction embedded in tool output, because it read sensitive material and then shared it, or because it performed an external action without authorization. Risk accumulates across turns and actions. Figure 1 contrasts three computations over the same trajectory: a classifier maps turns directly to a score; an LLM judge emits a verdict and generated rationale; BayesTrace carries turn evidence through relation, mechanism, and harm nodes to the posterior in one graph.

Benchmarks and auditors such as TRACES (Li et al. 2026a), AgentDoG (Liu et al. 2026), ATBench (Li et al. 2026b), R-Judge (Yuan et al. 2024), and ASSEBench (Luo et al. 2025) moved the field from single-turn text safety to

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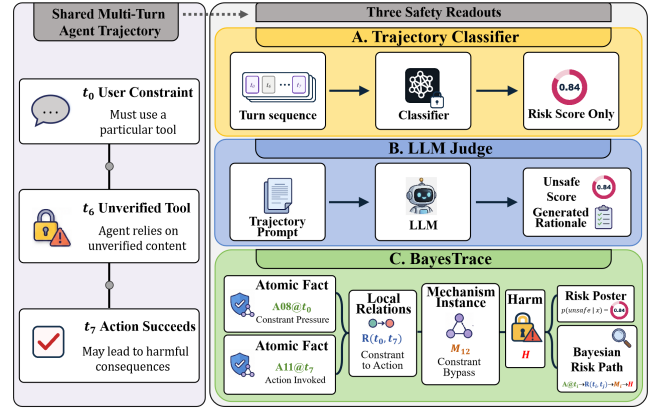


Figure 1: Three computations over the same trajectory.

trajectory-level detection and diagnosis. Their outputs are mostly trajectory-level labels, aggregate diagnoses, or free-form rationales. A black-box classifier can be accurate yet highlight the wrong turns; an LLM judge can produce a plausible rationale disconnected from the computation that produced its score; a flat concept model can find risk concepts while losing the temporal order and local relations that make them dangerous.

Simply drawing a graph around a classifier leaves this structure unresolved. A hand-fixed Bayesian graph is readable and applies the same active structure to every trajectory. A fully free graph learner adapts to each trajectory, yet its latent edges may lose a stable audit meaning and collapse toward direct predictors of the final label. The required model must satisfy four requirements at once: preserve uncertainty at every layer, score directed candidate dependencies between turns, instantiate different local chains for different trajectories, and expose the exact variables that carry the prediction.

We meet these requirements with a template-constrained dynamic Bayesian network. A finite global template space defines legal semantic roles, relation types, temporal windows, protection bindings, and harm links. For a specific trajectory, BayesTrace estimates turn-local atomic facts, scores candidate relations, separates their semantic probabilities from their sample-dependent influence, and instantiates only

the mechanism windows supported by that trajectory. All sample-dependent evidence contributions reaching the unsafe node must pass through mechanism instances and harm nodes; learnable leak terms provide only sample-independent priors. Raw text and atomic facts have no shortcut to the final decision. Together, these constraints define one layered probabilistic network whose conditional probabilities are parameterized by different learnable modules (Fig. 2).

The activated graph directly yields a Bayesian risk path. For a trajectory x , it is the ranked subset of the activated graph that connects concrete $A_i@t$ facts through $R_k@(s, t)$ relations, $M_j@w$ mechanism instances, and harm nodes to the risk posterior. It can be checked against reference labels and intervened on inside the model. The same representation supports local protection accounting. A protective act is credited only when it binds to the same mechanism instance, and an outcome-conditioned analysis can further test whether later execution evidence contradicts the claimed mitigation. The resulting path represents both how risk accumulates and whether a protective action interrupts the local risk chain.

Our contributions:

- **TraceAudit.** We normalize 4,364 trajectories from four public benchmarks under a shared turn-to-harm audit schema, enabling cross-source training and evaluation of evidence, relations, mechanisms, harms, and protection.
- **BayesTrace.** We introduce a template-constrained dynamic Bayesian network that instantiates a trajectory-specific subgraph and separates relation semantics from downstream influence.
- **Evaluation.** We jointly test risk detection, path-content alignment, and computational path use through matched path interventions.

Figure 2 previews the shared template space and the trajectory-specific subgraph; Section 5 defines their learning and risk-path readout.

2 Related Work

Agent safety benchmarks and trajectory auditors. ToolEmu (Ruan et al. 2024), AgentHarm (Andriushchenko et al. 2025), AgentDojo (Debenedetti et al. 2024), TrustAgent (Hua et al. 2024), Agent-SafetyBench (Zhang et al. 2024), and OS-Harm (Kuntz et al. 2025) show that tool use, external observations, and computer-use actions create safety failures beyond single-turn text moderation. TRACES (Li et al. 2026a) studies proactive monitoring over trajectory prefixes; AgentDoG (Liu et al. 2026) and ATBench (Li et al. 2026b) emphasize diagnosis taxonomies and guardrails; R-Judge (Yuan et al. 2024) and ASSEBench (Luo et al. 2025) provide risk-awareness and safety/security scenarios. TraceAudit targets a narrower problem: learning mechanism-instance-level Bayesian risk paths whose semantic content and computational role can both be tested.

Judges, guardrails, and post-hoc explanations. LLM judges and guardrail classifiers (Zheng et al. 2023; Inan et al. 2023; Han et al. 2024; DeepSeek-AI 2025) yield useful decisions, and control protocols can use model monitors to reduce harmful outputs (Greenblatt et al. 2024). These explanations

can diverge from the scoring computation: a judge’s rationale can cite turns that played no role in its verdict, and nothing in the scoring objective necessarily penalizes that mismatch. Our protocol separates two questions. Semantic alignment asks whether the highlighted turns, mechanisms, harms, and relations match reference audit objects. Computational path use asks whether intervention on the highlighted path moves the model’s own posterior more than matched controls. TraceAudit therefore evaluates both the activated-subgraph content and its role in the same probabilistic computation.

Concept bottlenecks and probabilistic graphical models.

Concept bottlenecks and self-explaining models (Koh et al. 2020; Alvarez-Melis and Jaakkola 2018) route predictions through named intermediate variables, while rationale and saliency work studies whether explanations align with evidence or affect the model (Jacovi and Goldberg 2020; DeYoung et al. 2020; Jain and Wallace 2019; Sundararajan, Taly, and Yan 2017; Hooker et al. 2019). These lines motivate our insistence that a risk path be both semantically checkable and load-bearing. Classical Bayesian networks and dynamic Bayesian networks (Pearl 1988; Koller and Friedman 2009; Murphy 2002; Heckerman 1989; Friedman, Geiger, and Goldszmidt 1997; Heckerman, Geiger, and Chickering 1995) give probabilistic semantics over structured variables, and differentiable structure-learning or neural-symbolic methods (Zheng et al. 2018; Yu et al. 2019; Manhaeve et al. 2018; Xu et al. 2018) show how structure and learning can be combined. BayesTrace uses these ideas differently: the global structure space is a small human-readable mechanism-template set, while each trajectory activates a sample-specific probabilistic subgraph inside that set.

3 Problem Definition

Given a trajectory $x = (u_1, \dots, u_T)$ where each turn u_t has a role (user, agent, environment), text, and optional tool metadata, and a label $y \in \{0, 1\}$ ($y=1$ unsafe), TraceAudit represents risk formation through five semantic layers. Atomic facts $A_i@t$ are observable claims localized to a turn, such as an external instruction, task adoption, an invoked action, a sensitive share, or a source check. Local relations $R_k@(s, t)$ state how two ordered turns participate in one mechanism, such as an external instruction shaping a later task interpretation. Mechanism instances $M_j@w$ bind the required facts and relations inside a concrete window w . Harm nodes H_h collect mechanisms with the same consequence, and Y is the trajectory-level unsafe variable.

Let $\mathcal{T}$ denote the globally shared mechanism-template space. It defines legal node roles, relation directions, temporal windows, protection bindings, and mechanism-to-harm links. A trajectory instantiates a subgraph $G(x) \subseteq \mathcal{T}$ whose node set depends on its turns and whose activation probabilities depend on its content. The required output is a calibrated posterior $p(Y=1 \mid x)$ and a Bayesian risk path $\pi_B(x) \subseteq G(x)$ containing the highest-contribution facts, relations, mechanism instances, and harms. Protective behavior and governance context are reported with their local binding status. The posterior-producing graph supplies $\pi_B(x)$

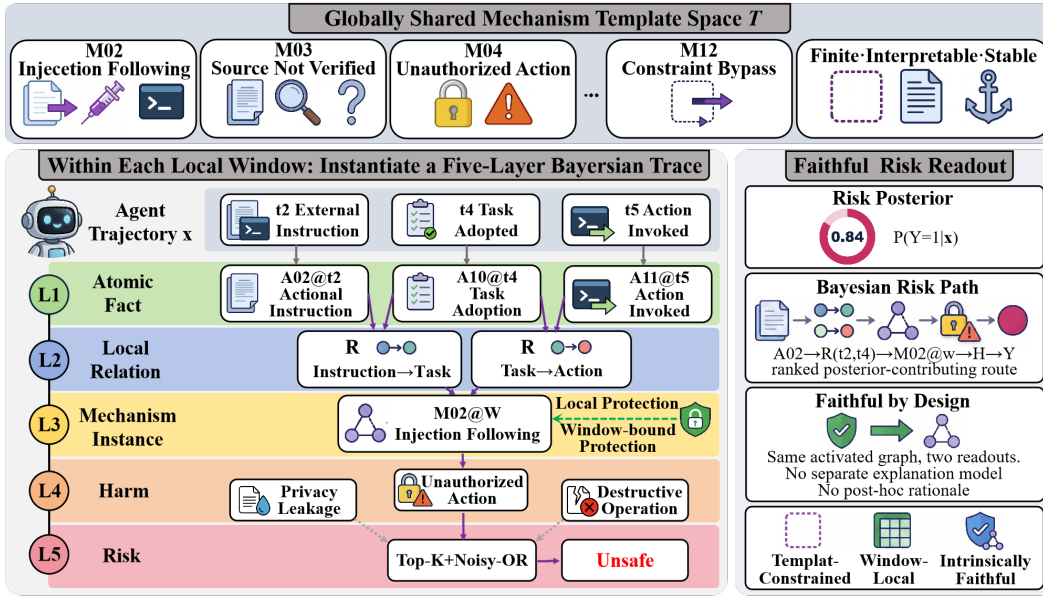


Figure 2: BayesTrace as a layered Bayesian network.

directly; no separate explanation model is used.

4 The TraceAudit Dataset

TraceAudit contains 4,364 trajectories (2,263 unsafe / 2,101 safe; mean 6.5 turns) from ASSEBench (2,293), ATBench (1,000), ATBench500 (500), and R-Judge (571). After source normalization, trajectories are pooled, randomly shuffled, and split approximately 60/20/20 into 2,616 / 871 / 877 train/validation/test examples.

Why a shared audit schema matters. The source benchmarks provide diverse scenarios and final safety targets, but they do not share a common vocabulary for how risk forms inside a trajectory. TraceAudit preserves each source label while overlaying the same atomic, relation, mechanism, and harm schema across all sources. This common layer turns heterogeneous outcome examples into comparable audit objects, supports shared intermediate supervision, and enables layer-wise evaluation across datasets.

All reported thresholds and the final relation configuration are frozen on validation. The test split is used for reported comparisons; outcome-status annotations used by the protection analysis are restricted to training and validation.

4.1 Label Hierarchy

We denote the annotation and target layers by A (atomic facts), R (local relations), M (mechanisms), H (harm categories), and L (the final trajectory label). L is inherited from each source benchmark as its safe/unsafe target and is represented by Y in the model equations. A, R, and M are produced through an LLM-assisted, five-annotator consensus process. Five annotators independently review identical candidates; a sample-object label is retained when at least three select the same label, and cases without a majority remain unannotated. Table 1 summarizes the two supervised

Table 1: Atomic-fact and mechanism-label categories.

Family	Unit	Main categories
A	turn	risk evidence; protective and governance behavior; action and outcome status
M	local window	injection following; unverified source; unauthorized action; leakage; destructive operation; harmful goal; constraint bypass; escalation and composition

label families; Supplementary Appendices A and C give the complete atomic codebook and mechanism template space.

H is not separately annotated. Each confirmed mechanism is mapped to one or more harm categories through the fixed, human-defined mechanism-to-harm ontology; these mapped categories form the H reference objects used in evaluation. The primary BayesTrace model uses A01–A22, while A23/A24 support the targeted protection-outcome analysis described next.

4.2 Protection-Outcome Annotation Slice

To evaluate whether a protective action actually interrupts a risky mechanism, we annotate two additional outcome facts on a targeted protection-analysis slice. A23 marks an agent turn that dispatches or performs an action. A24 marks an environment or tool turn that objectively confirms success; an agent’s own completion claim never counts as A24.

A five-annotator review of 138 trajectories produced 1,459 consensus labels under the same majority rule. We calibrated an LLM labeling pipeline on this gold set and used it to annotate roughly 7,000 candidate turns on the training and validation splits. On held-out consensus turns, the pipeline reaches F1 0.930 for A23 and 0.970 for A24. The full protocol — role conventions, prompt iterations, exclusion of tool-prefilled re-

views, and the explicit-only supervision mask used to train the extractor heads — is documented in Supplementary Appendix B.

5 Method: BayesTrace

BayesTrace learns Bayesian risk paths within one layered network. Different modules parameterize the conditional probabilities of atomic, relation, mechanism, harm, and risk variables, while the template space constrains which conditional dependencies are legal (Figs. 1 and 2).

5.1 Why a Template-Constrained Bayesian Network?

The architecture follows from four requirements. First, uncertain evidence must remain probabilistic: a low-confidence instruction should not become a hard fact before later reasoning. Second, risk depends on directed local dependencies in addition to the presence of concepts anywhere in the trace. Third, trajectories differ in length and evidence placement, so each sample must instantiate a different local graph. Fourth, every variable used for the decision must retain a stable risk-path meaning. A fixed graph preserves semantic roles and reuses one active structure. A free latent graph adapts to each sample without guaranteeing stable semantics. BayesTrace combines a fixed semantic template space with learned sample-specific activation.

Structure learning therefore occurs at three levels. Human and model-assisted design defines the legal template space $\mathcal{T}$, which acts as a structural prior. Training learns relation retention, role weights, protection strengths, mechanism-to-harm links, and risk propagation parameters inside that space. At inference, the current trajectory determines which candidate relations and mechanism windows activate. The graph forbids raw-text-to-risk, atomic-to-risk, and mechanism-instance-to-risk shortcuts: evidence must pass through mechanism and harm layers. This restriction makes the displayed risk path load-bearing for prediction and prevents a nominally structured model from collapsing into a star-shaped classifier.

5.2 Atomic Facts and Dynamic Local Relations

Each turn embedding e_t (frozen MiniLM (Wang et al. 2020) encoder) feeds per-label heads $p(A_i@t) = \sigma(f_i(e_t))$, trained with masked BCE (missing annotations are not negatives). The primary model uses A01–A22; the protection-outcome analysis adds A23/A24 with explicit-only masks (Sec. 4.2). For each legal relation template R_k and ordered pair (s, t) , the graph first computes a structural conditional

$$r_{kst}^{\text{str}} = g_k \sigma((w_k^s a_s + w_k^t a_t + w_k^\Delta \phi(t-s) + b_k) / \tau_k),$$

where a_s and a_t collect role-compatible atomic probabilities and $\phi(t-s)$ encodes distance. Learnable retention $g_k = \sigma(\gamma_k)$, temperature τ_k , and role weights determine whether a candidate is useful inside a mechanism window. A pairwise head additionally estimates the semantic conditional $u_{kst}^{\text{sem}} = \sigma(h_k(e_s, e_t))$ from the two endpoint embeddings; it is pretrained with the available pair annotations. The explicit risk-path node is $r_{kst}^{\text{sem}} = g_k u_{kst}^{\text{sem}}$.

Relation meaning and downstream influence are distinct quantities. Let $p_0(x)$ be the coarse posterior obtained from structural conditionals and let $c(x) = 1 - \exp[-(|p_0(x) - \theta_0|/\delta)^2]$. For an aligned semantic head, $\eta_k(x) = \alpha_k c(x)$ and the probability consumed by mechanisms is

$$r_{kst}^{\text{eff}} = g_k \sigma\left((1 - \eta_k) \logit \frac{r_{kst}^{\text{str}}}{g_k} + \eta_k \logit u_{kst}^{\text{sem}}\right).$$

Other legal relation types use $r^{\text{eff}} = r^{\text{str}}$. The path readout reports r^{sem} , $c(x)$, and the effective contribution entering the mechanism layer. The same graph evaluation produces all three variables; no separate explanation model is involved. Sparsity discourages unsupported relation families without imposing a fixed display threshold.

5.3 Mechanism Instances and Local Protection

Each template specifies arity, roles (required atomic label sets), relation constraints, window size, protective labels, and harm types. For candidate window w , $E_j(w)$ contains both its required atomic facts and its effective relations, and $q_j(w) = \sigma(\alpha_j \logit(\text{softAND}(E_j(w))) + b_j)$. Consequently, the same facts can yield a high mechanism probability when they form the required chain and a low probability when they occur in unrelated turns. Protection is deliberately local: A17 mitigates M03 only through the verification relation bound to the same external-content window; A18/A21/ A22 mitigate M04 only near the action window; A19/A20 are governance context by default. Effective probability: $\tilde{q}_j(w) = q_j(w) (1 - \rho_j P_{\text{prot}}(j, w))$ with learnable strengths ρ_j .

Outcome-conditioned protection. BayesTrace can optionally condition local protection on observed outcomes. Within a matched action window, externally confirmed execution reduces the mitigation credited to refusal or block evidence. For an action candidate at turn u :

$$P_{\text{prot}}^{\text{eff}}(u) = P_{\text{prot}}(u) \cdot \left(1 - \sigma(g) \max_{v \in [u, u+2]} p(A24@v)\right). \quad (1)$$

An analogous trace-level readout credits refusal only when no later success is confirmed (Supplementary Appendix B). The local and trace-level gate strengths initialize at $\sigma=0.5$ and are trained by the final risk loss. Their effects remain visible in the path output: it exposes the protective claim, the contradicting outcome, and the resulting mitigation strength.

5.4 Bayesian Propagation, Learning, and Risk-Path Readout

For each mechanism type, only the top- K supported instances enter a noisy-OR aggregation. Template-masked mechanism-to-harm parameters β_{jh} and harm-to-risk parameters ω_h propagate uncertainty as

$$z_h = 1 - (1 - \ell_h) \prod_{j,w} (1 - \beta_{jh} \tilde{q}_j(w)), \quad (2)$$

$$p(Y=1 | x) = 1 - (1 - \ell_Y) \prod_h (1 - \omega_h z_h), \quad (3)$$

where ℓ_h and ℓ_Y are learnable, sample-independent prior terms. They encode background prevalence and remain independent of trajectory text. The masks preserve the global semantic structure; factors inside each product depend on sample-specific instances.

The end-to-end objective is

$$\mathcal{L} = \mathcal{L}_Y + \lambda_A \mathcal{L}_A + \lambda_M \mathcal{L}_M + \lambda_S \mathcal{L}_{\text{sparse}} + \lambda_L \mathcal{L}_{\text{logic}} + \lambda_R \mathcal{L}_{\text{rank}} + \lambda_N \mathcal{L}_{\text{neg}}.$$

Here $\mathcal{L}_A$ is the turn-level atomic auxiliary loss and $\mathcal{L}_M$ is the direct mechanism-classification loss; $\mathcal{L}_{\text{rank}}$ and $\mathcal{L}_{\text{neg}}$ use the same mechanism targets for ordering and negative control. The corresponding weights are 0.015, 0.02, 0.002, 0.02, 0.01, and 0.01. No edge/path label is added during the final graph-training stage. In that stage, $\mathcal{L}_M \rightarrow q_j(w) \rightarrow r_{kst}^{\text{eff}}$ backpropagates through the structural relation scorer. The gradient therefore continues through relations under direct mechanism supervision. Mechanism targets alone underdetermine relation type and endpoints; the relation-pretrained semantic head provides local calibration. Validation-frozen α_k , θ_0 , and δ control its influence on the final graph. The decision and display thresholds are calibrated on validation data. Display thresholds do not affect the risk decision. Learning a Bayesian risk path estimates the conditional probabilities and propagation strengths that select and weight legal $A \rightarrow R \rightarrow M \rightarrow H \rightarrow Y$ routes for each trajectory. The template space bounds the topology search. The risk-path readout ranks mechanism instances by their contribution to the posterior and expands each selected instance back to its supporting relations, turn facts, protection assessment, and harm node.

6 Experiments

We organize the evaluation around three research questions (RQs). RQ1 tests risk-detection performance, RQ2 isolates the role of each Bayesian component, and RQ3 evaluates whether the displayed risk path is both aligned with the reference content and used by the predictor.

6.1 Setup and Reading Guide

All model selection, thresholds, and label mappings are frozen on validation data. The main detection row and all RQ3 comparisons use the A01–A22 configuration with its validation-frozen semantic relation calibration and R-to-M influence configuration. A23/A24 appear only in the targeted protection-accounting analysis and its matched component controls, whose exact configurations are reported in Supplementary Appendix E. The merged split contains 2,616/871/877 train/validation/test trajectories.

Risk detection is evaluated on all 877 test trajectories. Risk-path content evaluation uses the 444 unsafe test trajectories containing at least one reference path object. Of these, 442 contain reference turns, mechanisms, harm categories, and atomic types, while 298 contain a positive reference relation annotation. We compute a set F1 per trajectory and then macro-average over trajectories with non-empty gold for that object. Turn-evidence F1 compares cited turn sets,

Table 2: Risk-detection performance on the test set.

Model	F1	AUROC	AUPRC	Acc
DeepSeek-V3.2 judge	0.595	0.699	0.703	0.685
Llama Guard 3 1B	0.546	0.500	0.518	0.517
TRACES-style GRU	0.843	0.911	0.911	0.826
AgentDoG uni-4b	0.806	0.830	0.801	0.803
BayesTrace	0.847	0.910	0.914	0.828

mechanism F1 compares mechanism-type sets, and harm-category F1 compares the H sets obtained from the fixed mechanism-to-harm ontology. A method is scored only for object types it emits. Stricter intrinsic diagnostics require the atomic type and turn or an exact (R_k, s, t) relation triple; their full definitions, denominators, and results are reported in Supplementary Appendix E.

External comparisons include a DeepSeek-V3.2 structured judge, an in-repository prefix-risk GRU inspired by TRACES, and eight completed AgentDoG 1.5 variants. Four AgentDoG variants emit taxonomy fields that can be evaluated for mechanism and harm alignment; the other four emit risk scores only. The TRACES-style localization rule (the three largest positive prefix-risk jumps) and the AgentDoG-to-TraceAudit taxonomy mapping are frozen on validation. Supplementary Appendix D gives training details and Appendix E gives complete baseline and per-source results.

6.2 Risk-Detection Performance (RQ1)

Table 2 reports performance on all 877 test trajectories. The BayesTrace row is the final reported checkpoint; repeated-training variation is analyzed below.

BayesTrace attains the highest point estimates in F1 (0.8467), AUPRC (0.914), and accuracy (0.828) in Table 2, while the TRACES-style GRU has the highest AUROC. Their paired F1 difference is 0.0036 with a 95% bootstrap interval of $[-0.0175, 0.0251]$ ($p = 0.739$). Across fixed-protocol training seeds 2029, 2030, and 2031, BayesTrace obtains 0.8397, 0.8458, and 0.8340 F1, respectively (0.8398 ± 0.0059). Thus BayesTrace adds structured Bayesian risk paths while retaining trajectory-level detection performance at the level of the strongest baseline.

6.3 Role of the Bayesian Structure (RQ2)

The Bayesian structure makes the intermediate risk path part of the prediction computation and gives each component a testable role. Table 3 uses controlled paired tests to examine complementary functions of the Bayesian structure. Each row evaluates the metric directly governed by the named component; arrows indicate the preferred direction, and comparisons are within rows.

Separating relation semantics from downstream influence yields 0.249 strict relation F1, compared with 0.000 when they share one control, while risk F1 is nearly unchanged (0.8467 versus 0.8458). The relation layer therefore supplies typed endpoints that mechanism targets alone do not determine. Direct mechanism supervision improves turn and mechanism content, sparsity reduces retained relation candidates, and logic strengthens path-deletion and risk-jump

Table 3: Functional tests of Bayesian components.

Test	Metric	Ref.	Variant
Couple R	Strict-R F1 $\uparrow$	0.249	0.000
No mech. sup.	Turn F1 $\uparrow$	0.214	0.161
No mech. sup.	Mech. F1 $\uparrow$	0.473	0.405
No sparsity	Retained R $\downarrow$	0.512	0.645
No logic	Path drop $\uparrow$	0.231	0.211
No logic	Jump loc. $\uparrow$	0.536	0.499

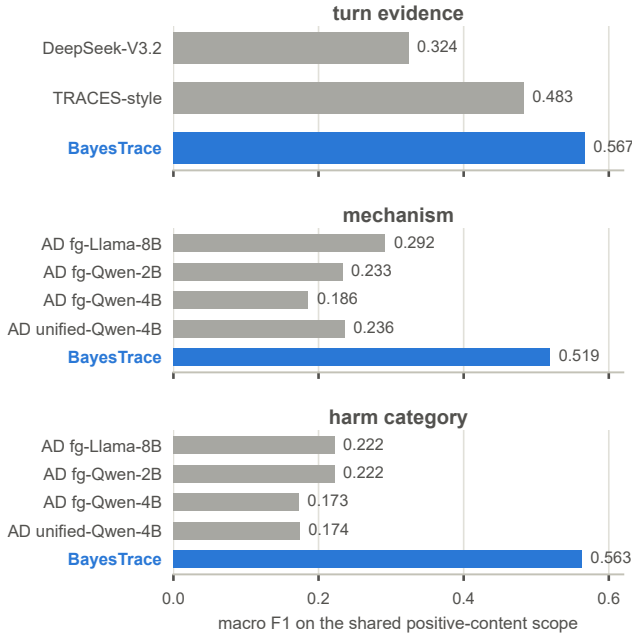


Figure 3: Risk-path alignment under the shared evaluation protocol. AD denotes AgentDoG 1.5.

tests. Together, the paired tests assign a distinct, measurable function to each Bayesian component. Full seed values are reported in Supplementary Appendix E.

6.4 Risk-Path Alignment and Computational Use (RQ3)

Content alignment and inspectability. At the turn, mechanism, and harm levels, the displayed content aligns with the available reference objects, and individual relations are exposed explicitly. Figure 3 makes this layered result explicit. When the atomic type is ignored and only the cited turn set is compared, BayesTrace reaches 0.567 F1, 8.35 points above TRACES-style and 24.27 points above DeepSeek-V3.2. For higher-level content, BayesTrace reaches 0.519 mechanism F1 and 0.563 harm-category F1. Across the four audit-capable AgentDoG variants, the strongest scores are 0.292 mechanism F1 and 0.222 harm F1, both from fg-Llama-8B under the same frozen taxonomy mapping. Under this shared mapping, BayesTrace improves mechanism alignment by 22.7 points and harm alignment by 34.1 points, while its own path readout also localizes the supporting turns.

Figure 4 makes the output concrete. For this trajectory, a

reader can check three non-crossing route lanes. The top lane contains the retained (t_0, t_7) relation. The middle M04 lane combines A11/A12 within t_7 , where a cross-turn relation is inapplicable. The bottom lane remains in the model computation, but its (t_6, t_7) relation candidate lies below the frozen display threshold and is rendered as a conditional gray path. All three lanes end at named harm categories and the same posterior. This is the practical audit object: the model exposes what it used, where it found it, and how it propagated through the graph. The visualization illustrates what the aggregate layer scores mean; aggregate evaluation supplies the quantitative evidence. Stricter atomic-type and exact-relation diagnostics are reported in Supplementary Appendix E; per-source localization results are reported in Supplementary Fig. 3.

Computational dependence. Content alignment alone does not show that the predictor used the displayed path. We therefore delete the turns supporting the highest-contribution path, recompute the posterior, and compare the result with deleting the same number of random or non-path turns. If the displayed risk path is load-bearing, its deletion should change the model’s own score more.

The result is positive: deleting BayesTrace’s selected risk path lowers its posterior by 0.2485 on average, compared with 0.1060 for random turns and 0.0489 for non-path turns (Fig. 5). The risk-path effect is therefore $2.34\times$ and $5.08\times$ the two controls. Posterior-jump localization is 0.593: in 59.3% of evaluated cases, the largest prefix-risk increase occurs inside the selected mechanism window. Together, these intervention results show that BayesTrace’s displayed risk paths are not post-hoc summaries: they identify the trajectory content that most strongly drives the model’s posterior.

7 Discussion

Why the Bayesian network is the contribution. The central object in BayesTrace is the conditional-dependency structure through which detection is computed. Atomic probabilities preserve uncertainty; semantic relation nodes identify directed candidate dependencies; sample-dependent influence controls their contribution to mechanisms; and harm nodes form the final aggregation boundary. A black-box backend with a post-hoc path generator lacks this shared computation.

The relation factorization changes model behavior. A mechanism instance receives both its required facts and their effective relations. The coupled-R ablation preserves the checkpoint and removes this separation. Its large strict-relation loss with almost unchanged risk F1 shows that final-label accuracy leaves relation usability untested. Positive-edge evaluation is therefore paired with the architectural constraint.

What is prior and what is learned. The template vocabulary is an explicit inductive bias for constrained structure learning. It fixes legitimate risk-path node types. Pair annotations pretrain local semantic conditionals; final risk and mechanism objectives learn structural relation scores and downstream propagation; validation freezes the semantic influence and display calibration. Component ablations

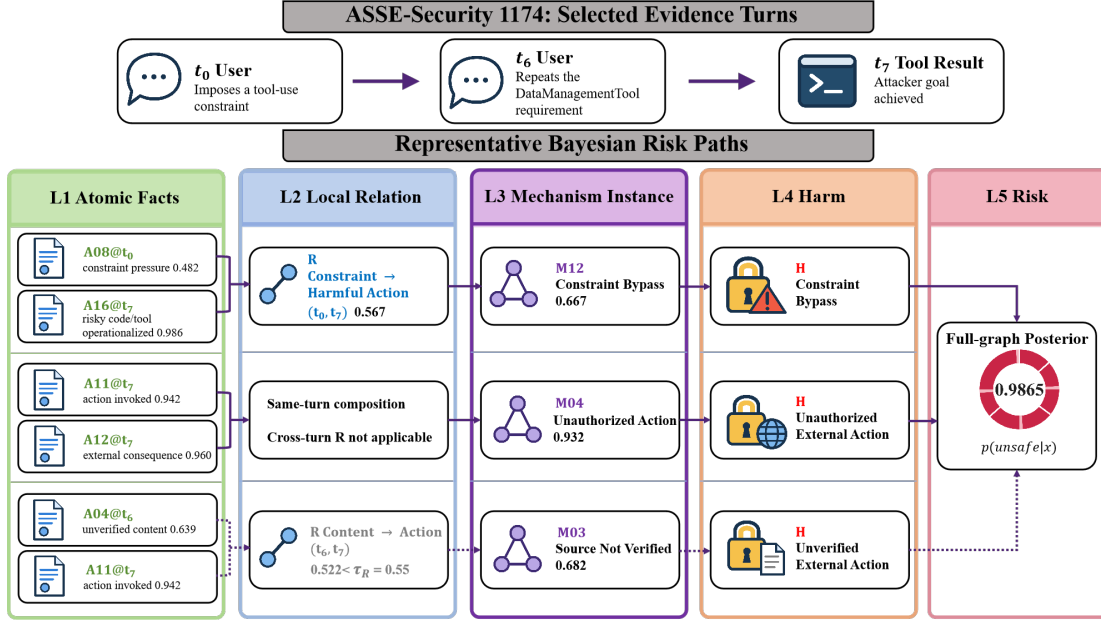


Figure 4: Representative Bayesian risk paths for an unsafe ASSE-Security trajectory. Solid purple edges denote displayed paths, while dotted edges mark a relation candidate below the fixed display threshold. The posterior is computed from the full activated graph.

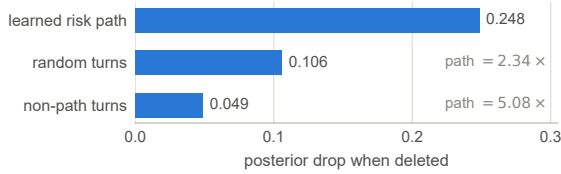


Figure 5: Mean posterior decrease after deleting matched turn sets.

show that direct mechanism supervision supports typed evidence and mechanism content, sparsity controls how indiscriminately candidates are retained, and logic supports load-bearing path tests. None of these results constitutes unrestricted causal discovery.

Semantic correctness and computational path use. Content alignment asks whether the model names the right path objects; deletion asks whether it used what it displayed. Together they show that the risk score and displayed path come from the same activated subgraph. The factorized relation readout further exposes endpoint semantics as an independently testable object.

8 Limitations

BayesTrace learns within a human-defined mechanism-template space; its probabilities describe model-internal dependencies, not unrestricted real-world causality. The current study evaluates covered mechanism families under a mixed-source split; expanding the template inventory and testing source-held-out transfer are natural next steps.

9 Ethical Statement

TraceAudit contains unsafe behavior; source licenses and redaction will be preserved, and annotations are not ground truth for high-stakes decisions. Generative AI assisted brainstorming, code and manuscript preparation, and preliminary figure design. The authors verified all final content and take responsibility; experimental LLM use is documented in Sections 4 and 6.

10 Conclusion

We presented TraceAudit and BayesTrace, an atomic-first dataset and template-constrained dynamic Bayesian network for trajectory-specific risk paths. Across three seeds, risk detection remains competitive while audit-content alignment exceeds available comparisons. By making prediction and audit two readouts of the same activated subgraph, BayesTrace moves trajectory safety analysis beyond a verdict to a checkable probabilistic account of how risk forms.

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